INTRODUCTION TO INFORMATION GEOMETRY

Three Strategies and the Simplex

Week 3 · Lecture 1

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Three proportions, two independent coordinates

Write the population state as $\mathbf{p} = (p_A, p_B, p_C)$.

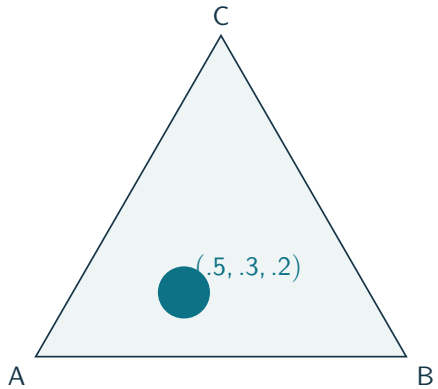
$$p_A + p_B + p_C = 1, \qquad p_i \geq 0.$$

The possible states form a triangle:

$$\Delta^2 = \{\mathbf{p} \in \mathbb{R}^3 : p_i \geq 0,\ \sum_i p_i = 1\}.$$

Its vertices are pure populations. An edge contains two types.

Drawing a distribution as a point



A state is the weighted average of the three vertex locations.

Barycentric coordinates

Choose planar vertices $A = (0, 0)$, $B = (1, 0)$, and $C = (\frac{1}{2}, \frac{\sqrt{3}}{2})$.

$$r(\mathbf{p}) = p_A A + p_B B + p_C C.$$

$$= \left(p_B + \frac{1}{2} p_C,\ \frac{\sqrt{3}}{2} p_C \right).$$

The triangle is an affine drawing of the normalization plane in $\mathbb{R}^3$.

Velocity is tangent to the normalization plane

If $\mathbf{p}(t)$ is differentiable, then

$$p_A(t) + p_B(t) + p_C(t) = 1.$$

Differentiate:

$$p'_A + p'_B + p'_C = 0.$$

An increase in one coordinate is balanced by decreases in the others.

The multi-strategy replicator equation

Suppose $N_i' = f_i N_i$ and $p_i = N_i / N$, where $N = \sum_i N_i$. The mean fitness is

$$\bar{f} = \sum_i p_i f_i.$$

The proportion equation for each type is

$$p_i' = p_i(f_i - \bar{f}).$$

Computing a three-type velocity

Let $\mathbf{p} = (0.2, 0.3, 0.5)$ and $\mathbf{f} = (3, 1, 2)$. Find the mean fitness and the vector $\mathbf{p}'$.

$$\bar{f} = \sum_i p_i f_i = ?, \qquad \mathbf{p}' = (\ ?,\ ?,\ ?).$$

Check that your components sum to zero.

Computing a three-type velocity

Let $\mathbf{p} = (0.2, 0.3, 0.5)$ and $\mathbf{f} = (3, 1, 2)$. The mean fitness is

$$\bar{f} = 0.2(3) + 0.3(1) + 0.5(2) = 1.9.$$

$$\mathbf{p}' = (0.22, -0.27, 0.05).$$

The components sum to zero.

Conservation of total probability

Sum the replicator velocities:

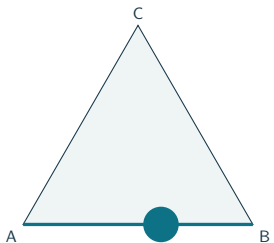
$$\sum_i p_i' = \sum_i p_i(f_i - \bar{f}) = \bar{f} - \bar{f}\sum_i p_i = 0.$$

Thus a solution starting in the normalization plane stays there. For smooth fitness functions, positivity of initially present types is also preserved at finite times.

The faces of a simplex are invariant

Whenever $p_C = 0$,

$$p'_C = p_C(f_C - \bar{f}) = 0.$$



On the AB edge, the system reduces to a two-strategy model.

Cyclic interactions

Consider a three-strategy payoff family:

$$M_s = \begin{pmatrix} 0 & 1 & -s \\ -s & 0 & 1 \\ 1 & -s & 0 \end{pmatrix}, \qquad s > 0.$$

$$f_A = p_B - s p_C, \quad f_B = p_C - s p_A, \quad f_C = p_A - s p_B.$$

A outperforms B, B outperforms C, and C outperforms A.

A cyclic velocity at one state

Take $s = 1$ and $\mathbf{p} = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$. Compute $\mathbf{f} = M_1\mathbf{p}$, then $\bar{f}$ and $\mathbf{p}'$.

$$\mathbf{f} = (\ ?,\ ?,\ ?), \qquad \mathbf{p}' = (\ ?,\ ?,\ ?).$$

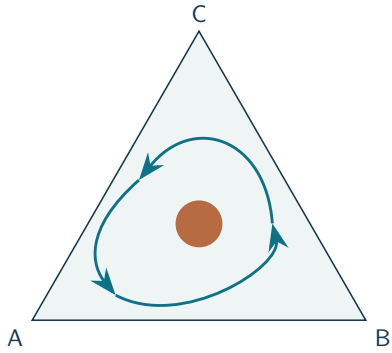
A cyclic velocity at one state

Take $s = 1$ and $\mathbf{p} = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$. The fitness vector is

$$\mathbf{f} = (\tfrac{1}{6}, -\tfrac{1}{3}, \tfrac{1}{6}), \qquad \bar{f} = 0.$$

$$\mathbf{p}' = (\tfrac{1}{12}, -\tfrac{1}{9}, \tfrac{1}{36}).$$

Three strategies permit turning



An interior trajectory has two independent velocity components. The cyclic model can circulate around the equal mixture $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

Antisymmetry and mean fitness

At $s = 1$, the payoff matrix satisfies

$$M_1^T = -M_1.$$

The scalar $\mathbf{p}^T M_1 \mathbf{p}$ therefore equals its own negative. Hence

$$\bar{f} = \mathbf{p}^T M_1 \mathbf{p} = 0$$

at every state, even though individual fitnesses drive motion.

Dimensions, support, and motion

For n outcomes,

$$\Delta^{n-1} = \{\mathbf{p} : p_i \geq 0,\ \sum_i p_i = 1\}.$$

Its dimension is $n-1$. The support consists of indices with $p_i > 0$. The next lecture uses the cyclic example to distinguish inward spiraling, closed orbits, and outward motion.